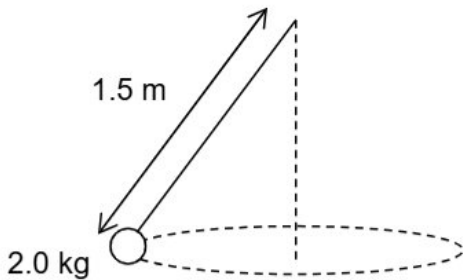
8 A conical pendulum consists of a mass of 2.0 kg attached to a light inextensible string of length 1.5 m. The string has a breaking tension of 30 N.



What is the maximum speed at which the mass can move in a horizontal circle before the string breaks?

A 3.6 m s^{-1} **B** 3.8 m s^{-1} **C** 4.1 m s^{-1} **D** 4.7 m s^{-1}